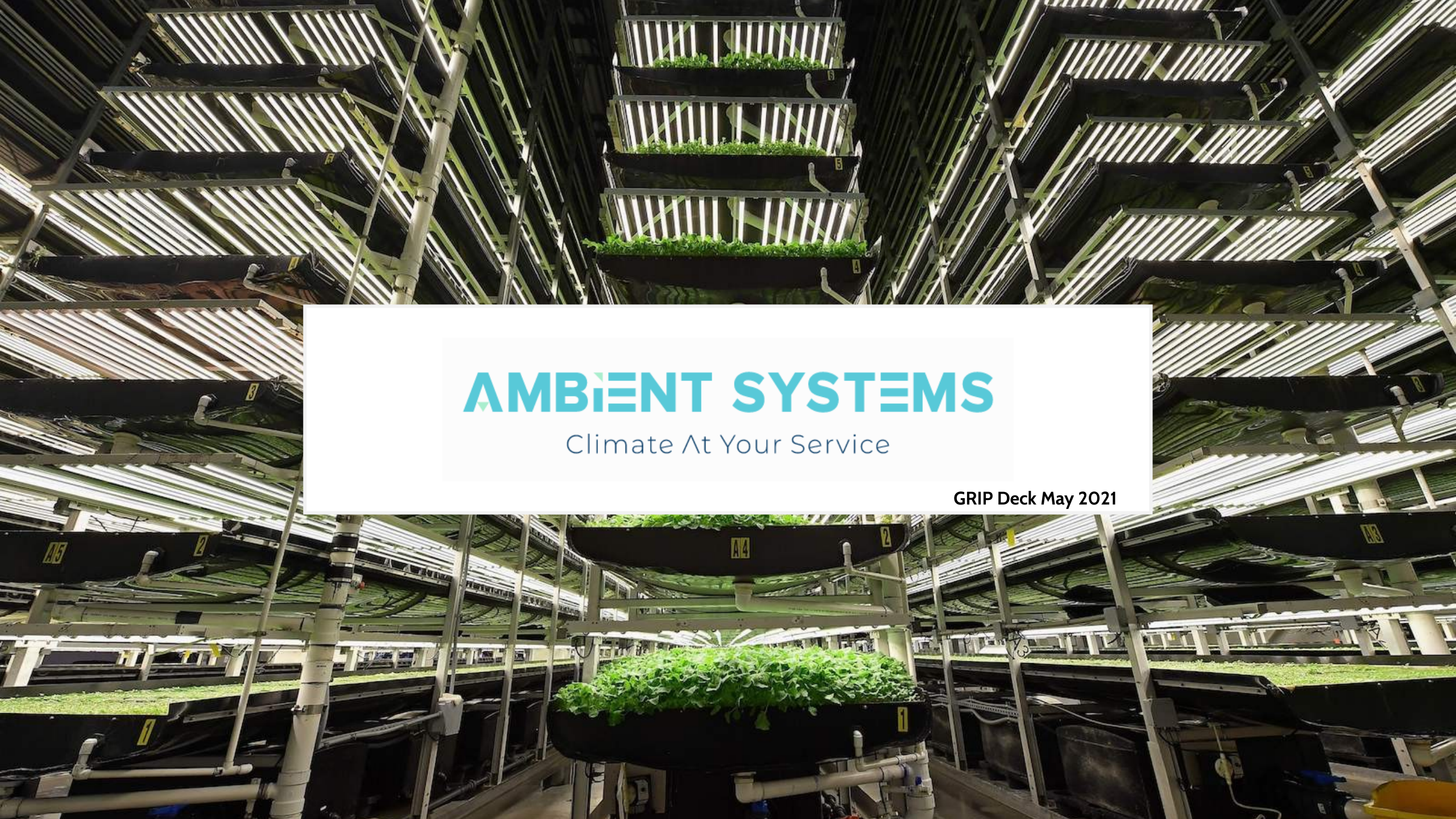


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Climate At Your Service

GRIP Deck May 2021

Building needs have becomes increasingly complex

Indoor Farms

Modern buildings with specific applications require more than

Healthcare

Restaurants

er

Getting the most out of operating a vertical farm

Utilities
$3



Rent
$30



Human Capital
$300

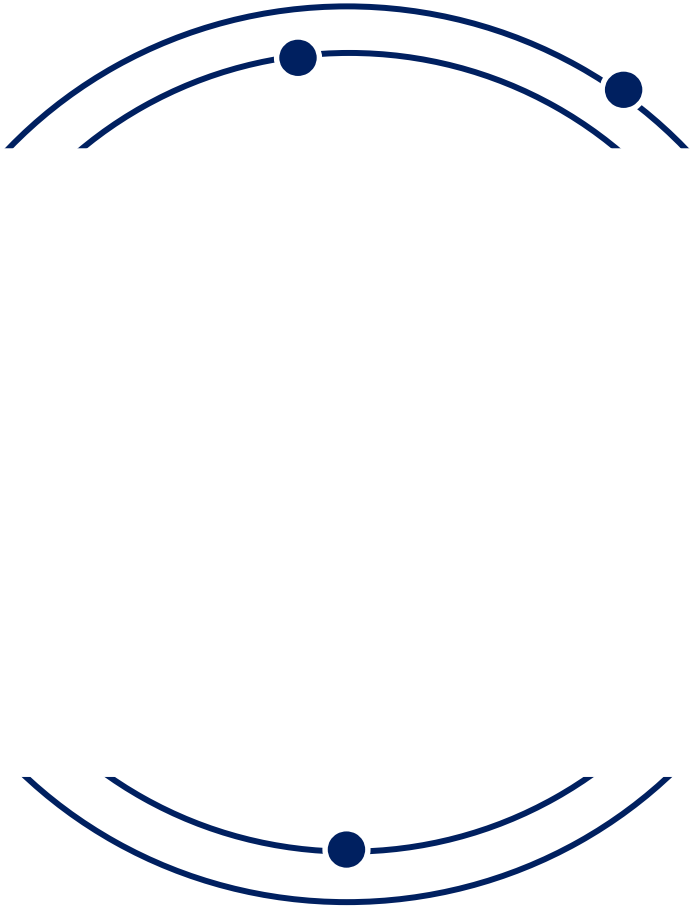


Source: Jones

Private and Confidentia

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Climate At Your Service

Climate Management Platform for Indoor Farms



Connected.

Centralized.

Automated.


A5
2
A4
2
A3
1
1

Understanding relationship between building conditions and crops

Temperature

Soil PH level

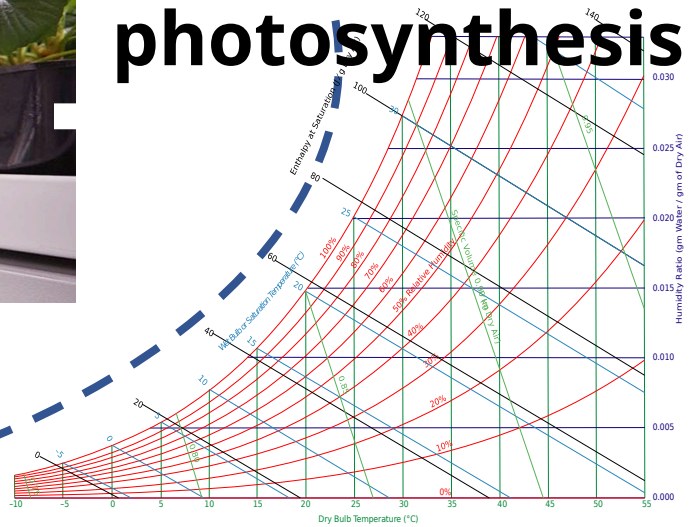
Lighting

Nutrients



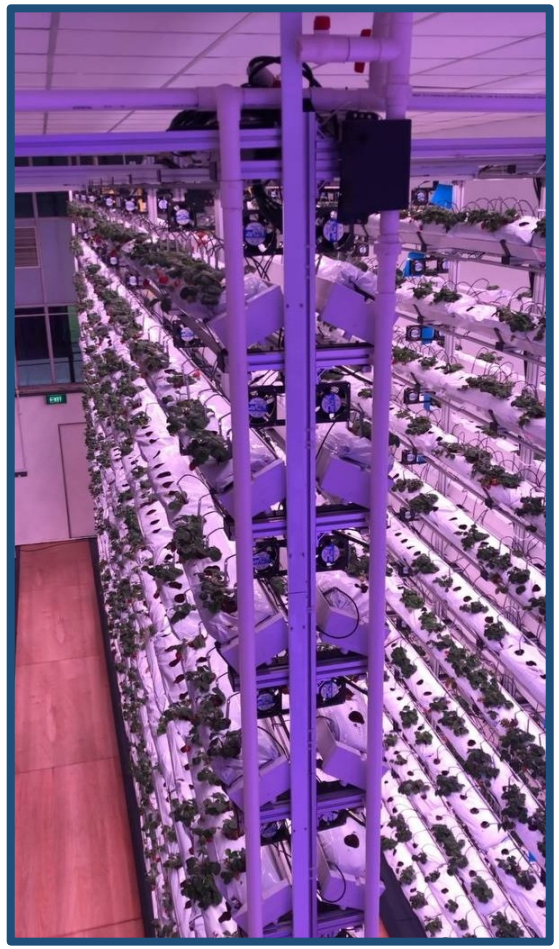
Humidity

Conditions outside of micro-ecosystem affect level of photosynthesis



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Undesired interactions between buildings and crops



Strawberries growing in varying levels of

Temperature
Humidity
CO2 Levels





Habitat for pests

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This is what Ambient will look like by end of 2021

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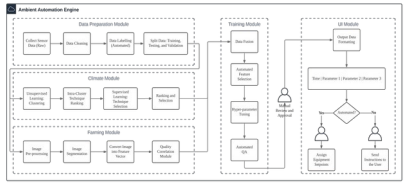
Current Product

Next 6 months

Sense → Optimize → Monitor → Controls



Smart Sensors



Optimization Engine

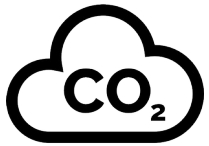
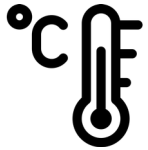


Dashboard



Edge Nodes & Control units

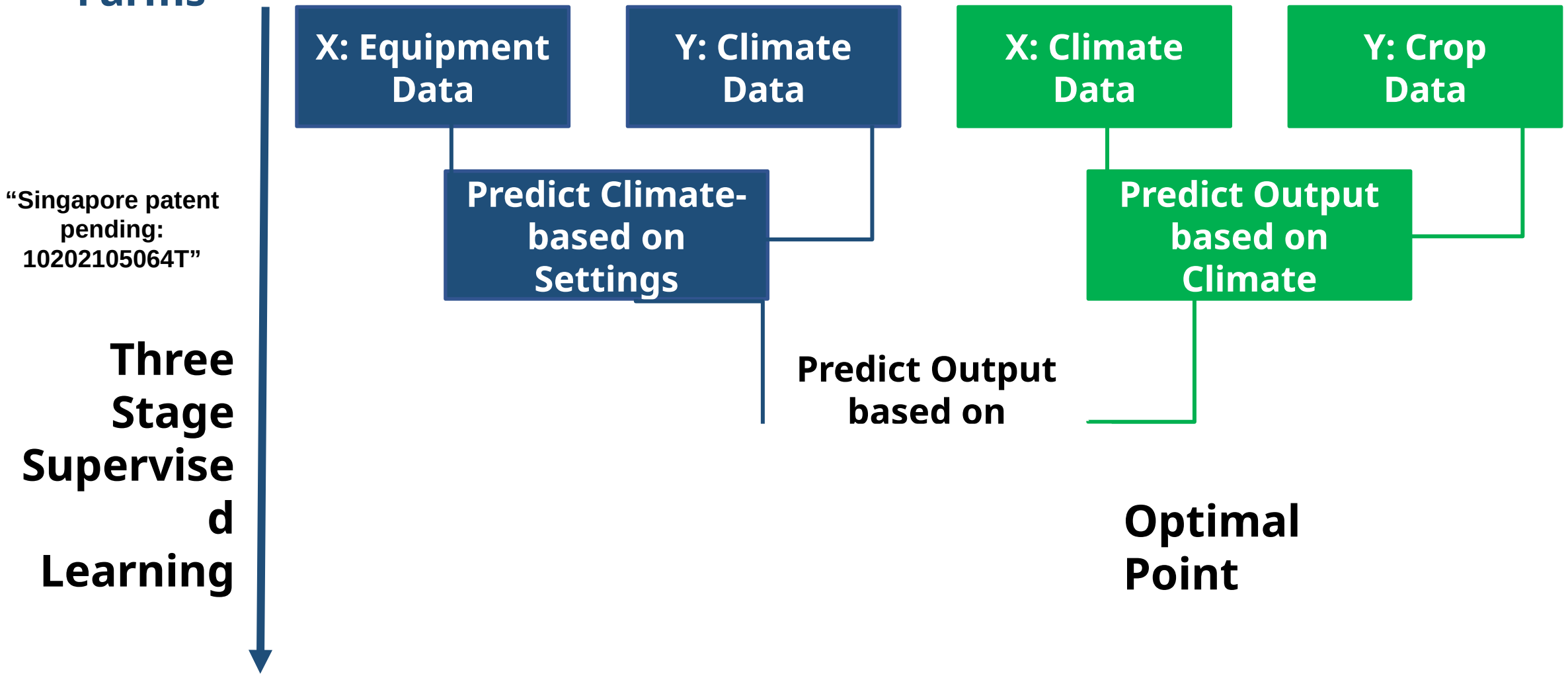
Insights from POC Pilot

Parameter	Finding	Potential Improvements
	CO2 at 25% of the optimal	• Increased photosynthesis by up to 75% • Earlier flowering • Higher fruit yields by up to 30% • Improved stem strength and flower size
	Temp 70% of the optimal	• Increased leaf and petiole growth • Brighter leaf color • Greater sucrose • More biomass at the roots

Ambient proprietary method provides equipment setpoints to achieve optimal targets

Time	Thermostat 1	Thermostat 2	Fan 1	Fan 2
May 28 – 2 pm	14 C	12 C	2 ACH	5 ACH

Patent: Computer System for Management of Buildings with Indoor Farms



"Singapore patent pending: 10202105064T"

Trade Secret: Operational Data on Creating an Optimal Indoor Climate

Climate Data

	Parameter
Current POC	Air Temperature
	Relative Humidity
	CO2
	Spectrum Light

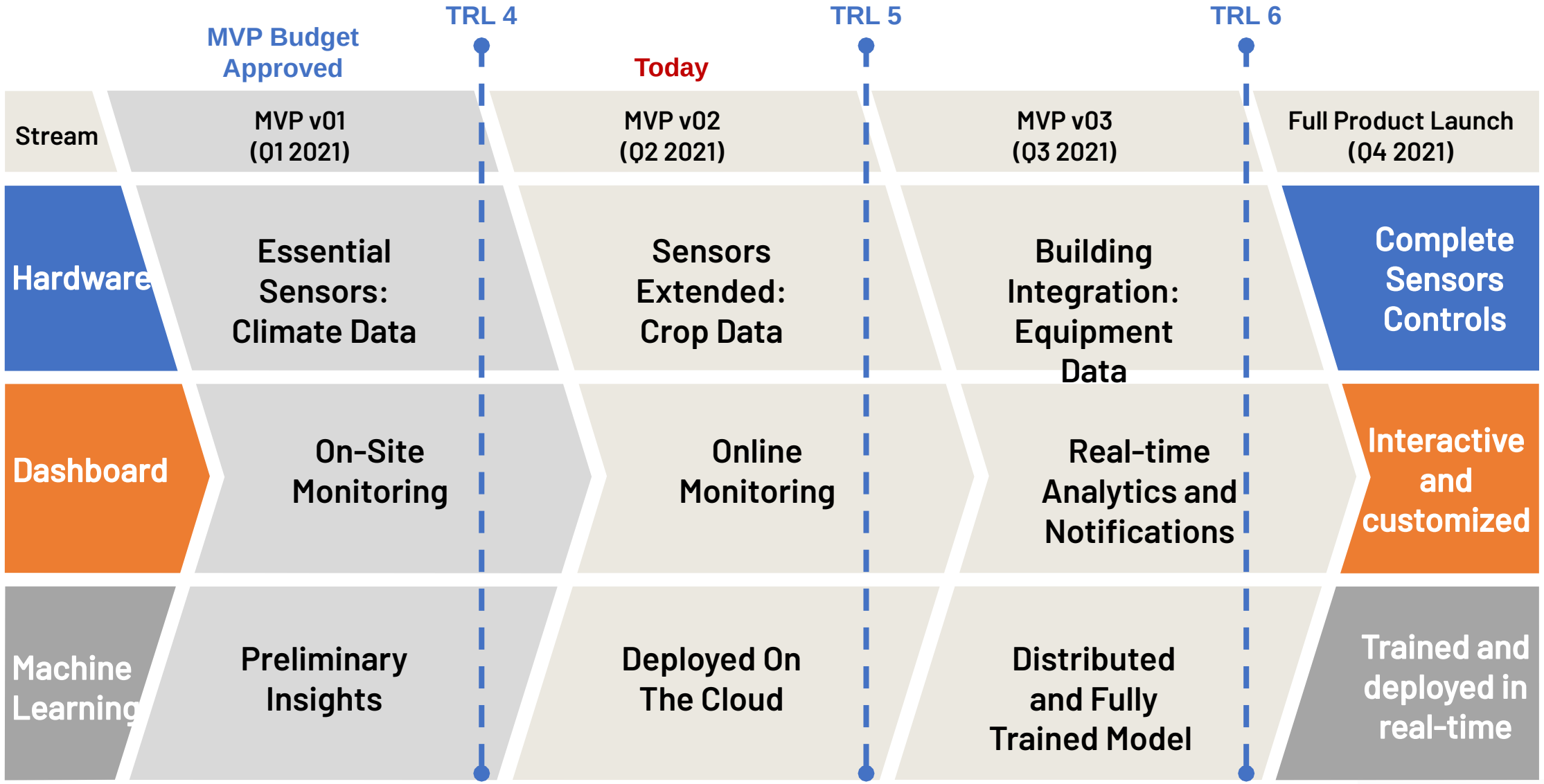
Crop Data

	Parameter
Full-scale Plot	Soil Temperature
	Soil Moisture
	pH level
	Electrical conductivity (EC_w)
	Crop specific parameters

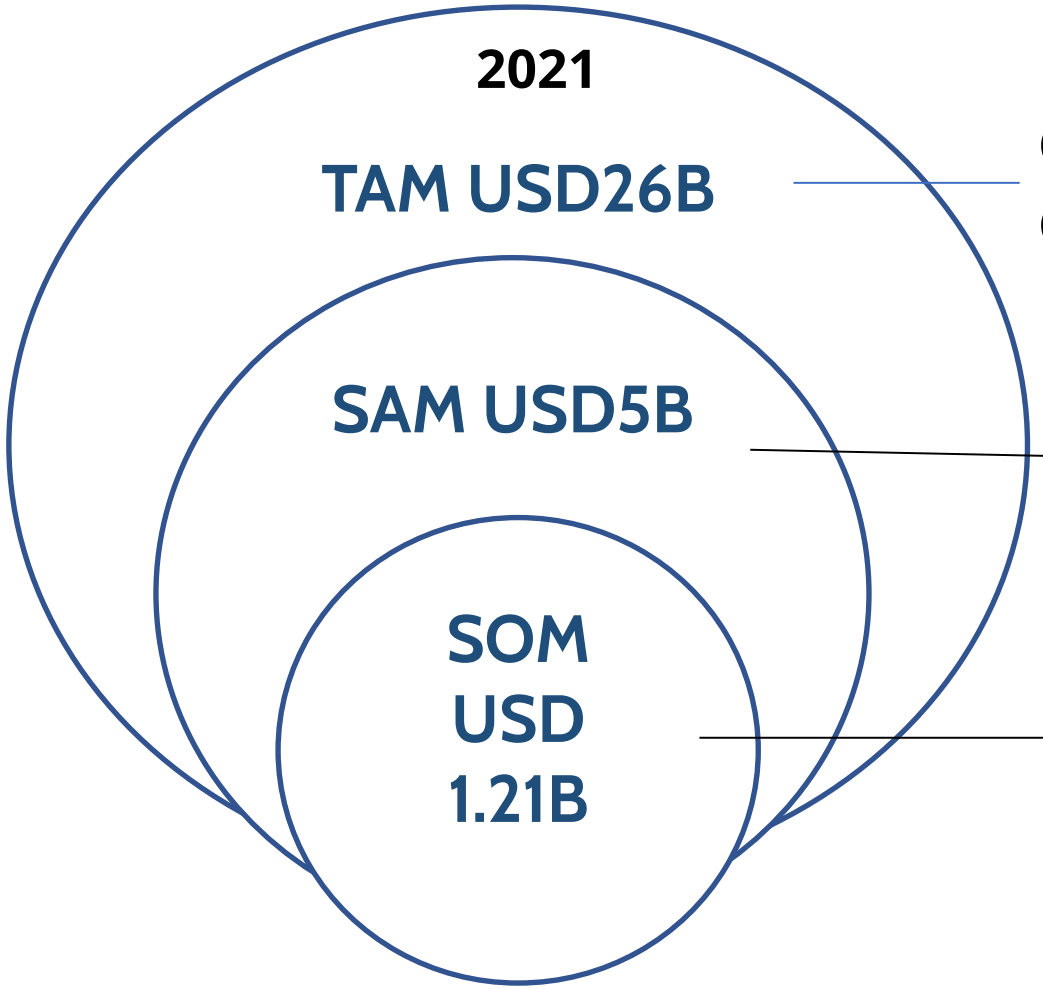
Equipment Settings

Sample Action
• Equipment setpoints • Light schedules • Chiller Plant settings • AHU Fan settings • Auxiliary Devices

Full product launch on-track by end of the year



Market Opportunity: TAM, SAM, SOM



Global HVAC Automation and Controls

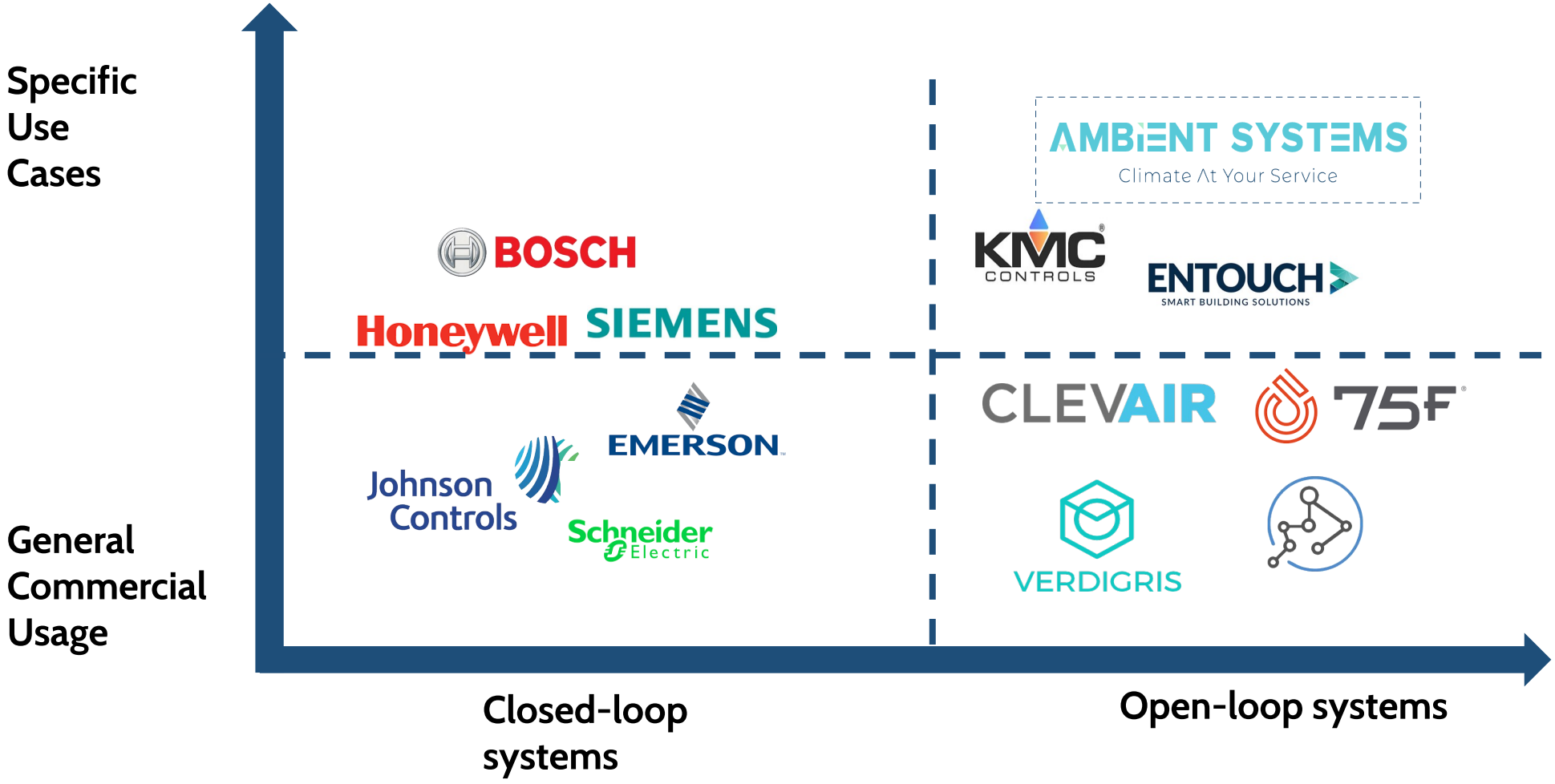
AI in Building Automation

APAC
Vertical Farming

Source: Frost & Sullivan, McKinsey, Data Bridge

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We provide value beyond building controls and energy savings



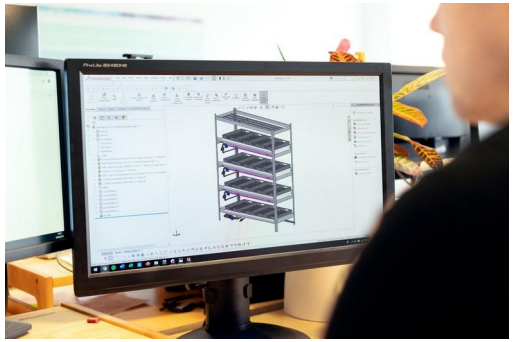
Source: Honeywell

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For farmers that are looking to collect and monitor building data

Yearly Subscription Model

Basic Monitoring



- **Real-time data**
- **Online dashboards**
- **Push Alerts**

One-time

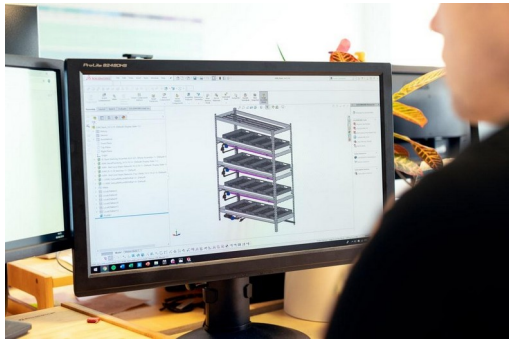
Installing Sensors
SGD1.12 / Sq ft

Per Month

Software Subscription
SGD500 / month

For farmers that are looking to automate their building conditions

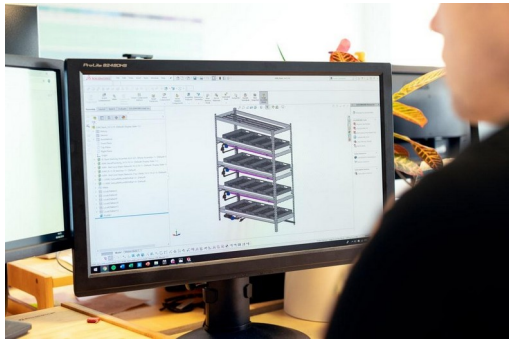
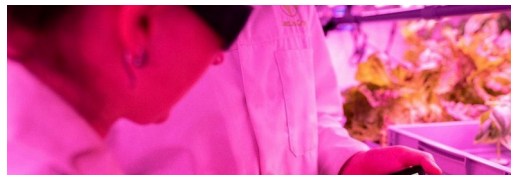
Yearly Subscription Model

	Basic Monitoring	Automating Building Controls
		
One-time	Installing Sensors SGD1.12 / Sq ft	Integration Cost SGD2.23 / Sq ft
Per Month	Software Subscription SGD500 / month	Cost of Maintenance $500 / month

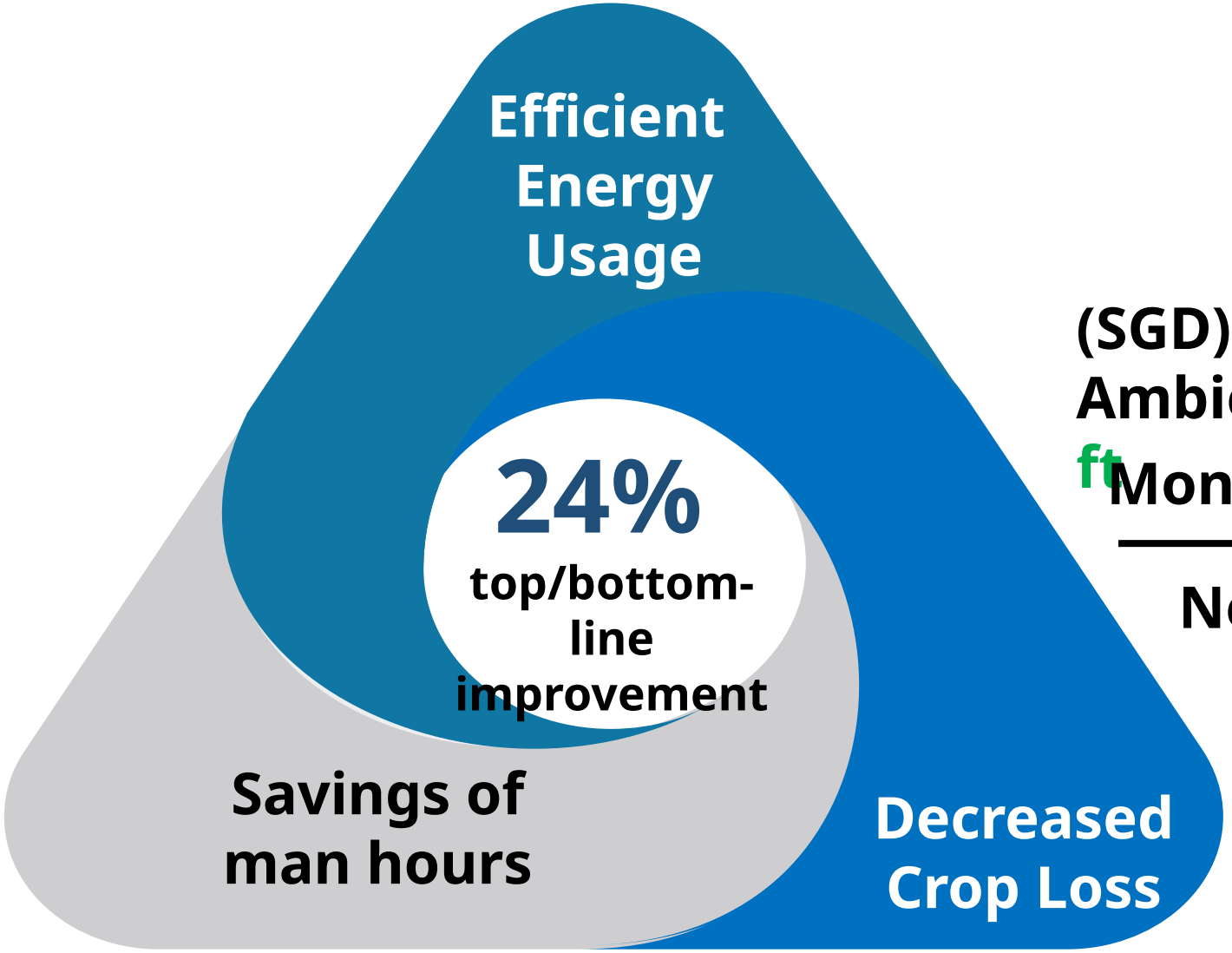
- **Full HVAC connectivity**
- **Real-time environmental adjustments**

For farmers that are looking to integrate crop and building data

Yearly Subscription Model

	Basic Monitoring	Automating Building Controls	Advanced Insights
			
One-time	Installing Sensors SGD1.12 / Sq ft	Integration Cost SGD2.23 / Sq ft	• Integrate user data • Obtain Actionable Insights according to needs
Per Month	Software Subscription SGD500 / month	Cost of Maintenance SGD500 / month	Additional Insights SGD300 / month

Farmers can expect to break-even installation costs in 4-6months



(SGD)

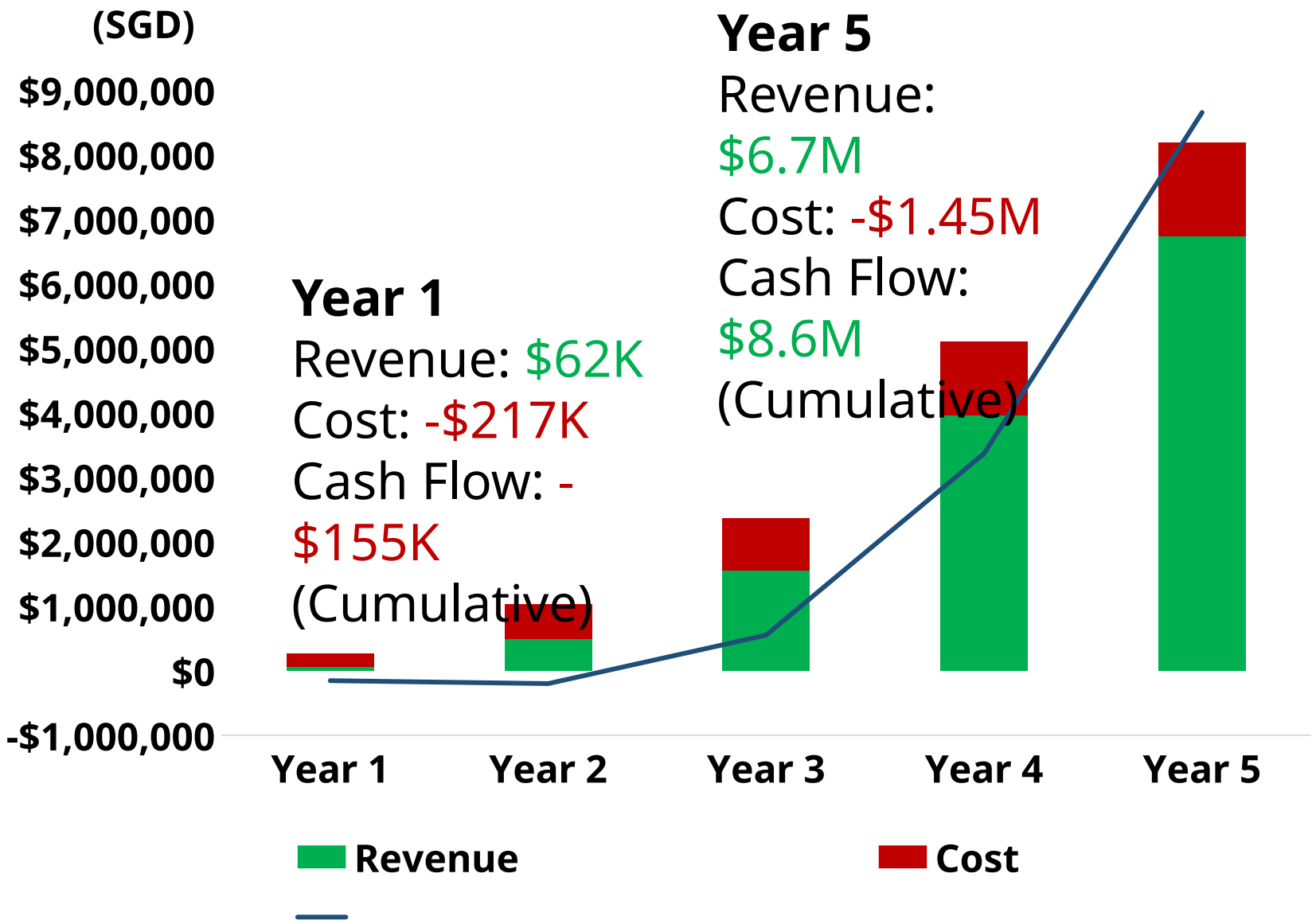
Ambient Value: $4.35 per sq ft

Monthly Cost: ($2.66 per sq ft)

Nett Value: $1.69 per sq ft

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Our financial projections

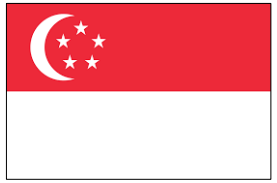


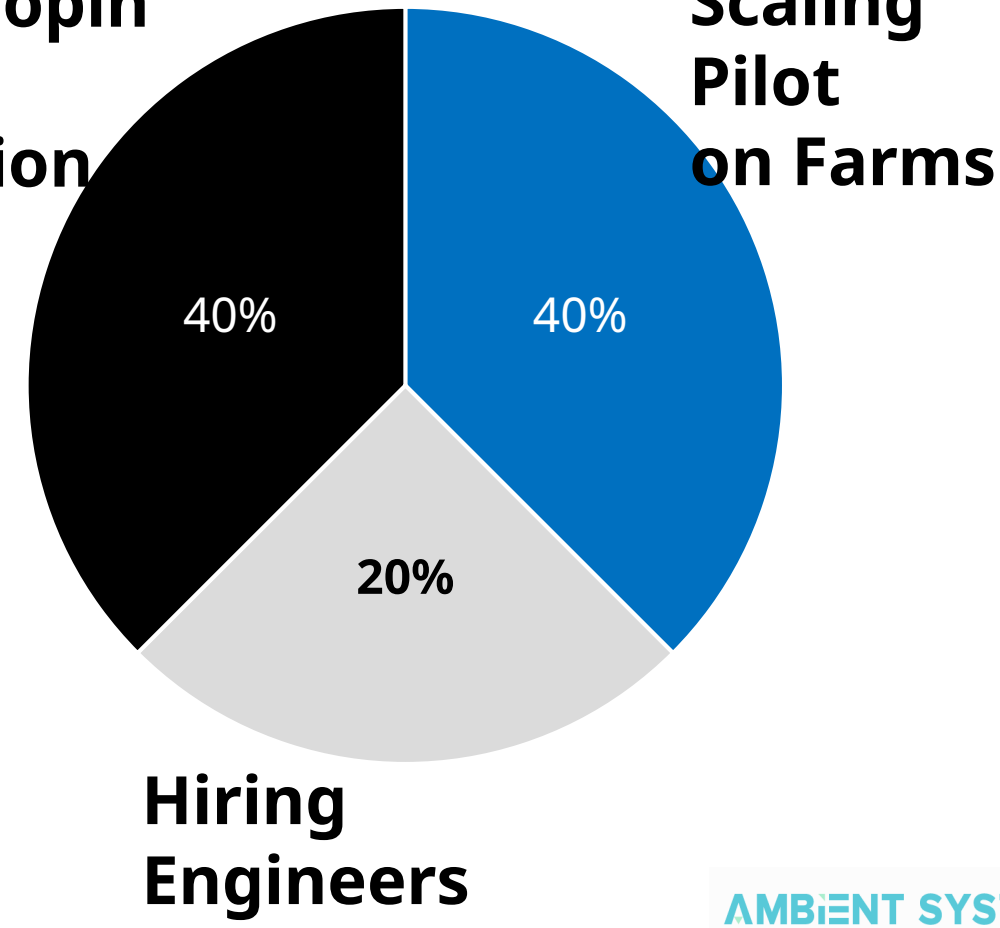
81.2%
Average Gross Profit

$8.5M
NPV

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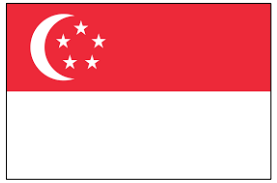
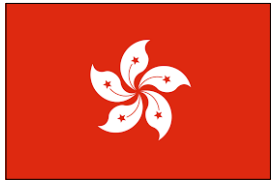
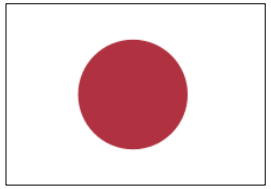
We have a funding gap of SGD252K over the next 18 months

	July 2021	Dec 2022
Market Expansion		
Offerings	Monitoring User Insights	
Segments	Farms	



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Aim to offer full-suite solution by Japan expansion

	2021	2022	2023	2024	2025
Market Expansion					
Offerings	Monitoring User Insights		Control		
Segments	Farms		Healthcare		Restaurants

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Received validation from farms

[CONFIDENTIAL]

Memorandum of Understanding

Comprehensive Climate Management System for Indoor Farming

Ambient Systems
Innovation 4.0
3 Research Link, #05-01
Singapore, 117602

Kingdom Farms
Singapore

November 5, 2020

Dear Mr. Timothy Tan,

Thank you for your interest in Ambient's climate management solution for indoor farms. At Ambient, we understand that managing the indoor environment can be a complex and time-consuming process. We want to make this easier for you by providing a comprehensive solution that handles all of your building management activities, allowing you to focus on your business.

As a deep-tech spin-off from National University of Singapore, Ambient's goal is to build solutions that will support urban farm operators with meeting their climate needs by leveraging technological advances in Internet-of-Things (IoT) and Machine Learning (ML). We propose to work together with Kingdom Farms on developing a climate management solution for your fleet of indoor farms. The motivation for this joint project is two-fold. First is to provide the necessary technology to effectively manage the indoor climate. Second is to leverage the collected data to develop a more precise method of indoor farming to achieve greater yield, higher quality, and more accurate harvest time predictions.

Scope of Work

Ambient's international team of engineers, software developers, and management consultants will help you obtain the most appropriate climate management solution for your needs.

The following services are offered as part of the overall solution:

Ambient Systems
i4 Building
3 Research Link #05-01
Singapore
117602

9th May 2021

Letter of intent

SINGROW
Growing Your Daily Fresh

Dear Ivan Damnjanovic,

I am writing to express our intention to do a pilot of Ambient's climate management solution at Singrow's indoor farm located in 102 Henderson Rd, Singapore 159562

Singrow is a Singapore-based company that develops innovative agritechnology solutions for producing premium-quality fruits and vegetables. Leveraging its proprietary platform, Singrow develops the freshest, best quality produce, such as non-GMO strawberry varieties in red and white aesthetics. More importantly, Singrow focuses its efforts on developing technologies that would help overcome ongoing challenges, such as global supply barriers, to meet the demand of the world's most discerning consumers.

Ambient understands that managing the indoor environment can be a complex and time-consuming process. Its mission is to make this process easier by providing a comprehensive solution that handles all building management activities, allowing indoor farm operators to focus on their core expertise in agritechnology.

Ambient proposes to work with Singrow on piloting a climate management solution at Singrow's indoor farm. The motivation for this project is two-fold. First is to provide the necessary technology to effectively manage the indoor climate. Second is to leverage the collected data to develop a more precise method of indoor farming to achieve greater yield, higher quality, and more accurate harvest time predictions.

We have received

2 LOI

from Kingdom Farms and SinGrow

Approx. Value for a 500 sq ft farm

SGD7.6K

AMBIENT SYSTEMS
Climate At Your Service

About Us

- **MComp, Computer Science**
  National University of Singapore
- **BASc, Civil Engineering, Sustainable Energy**
  University of Toronto
- **Serial Entrepreneur**
  5 years experience in cleantech, deep tech

, Finance
Singapore Management University

- **Strategy and Fundraising**
  3 years experience across
  e-commerce, gaming and payments

1. Experienced Team with tech, strategy and fundraising background

2. Received 2 LOI with indoor farms

3. Pilot test conducted with SinGrow resulted in actionable recommendations

4. Our Indoor Farming launch market growing at a CAGR of 24%

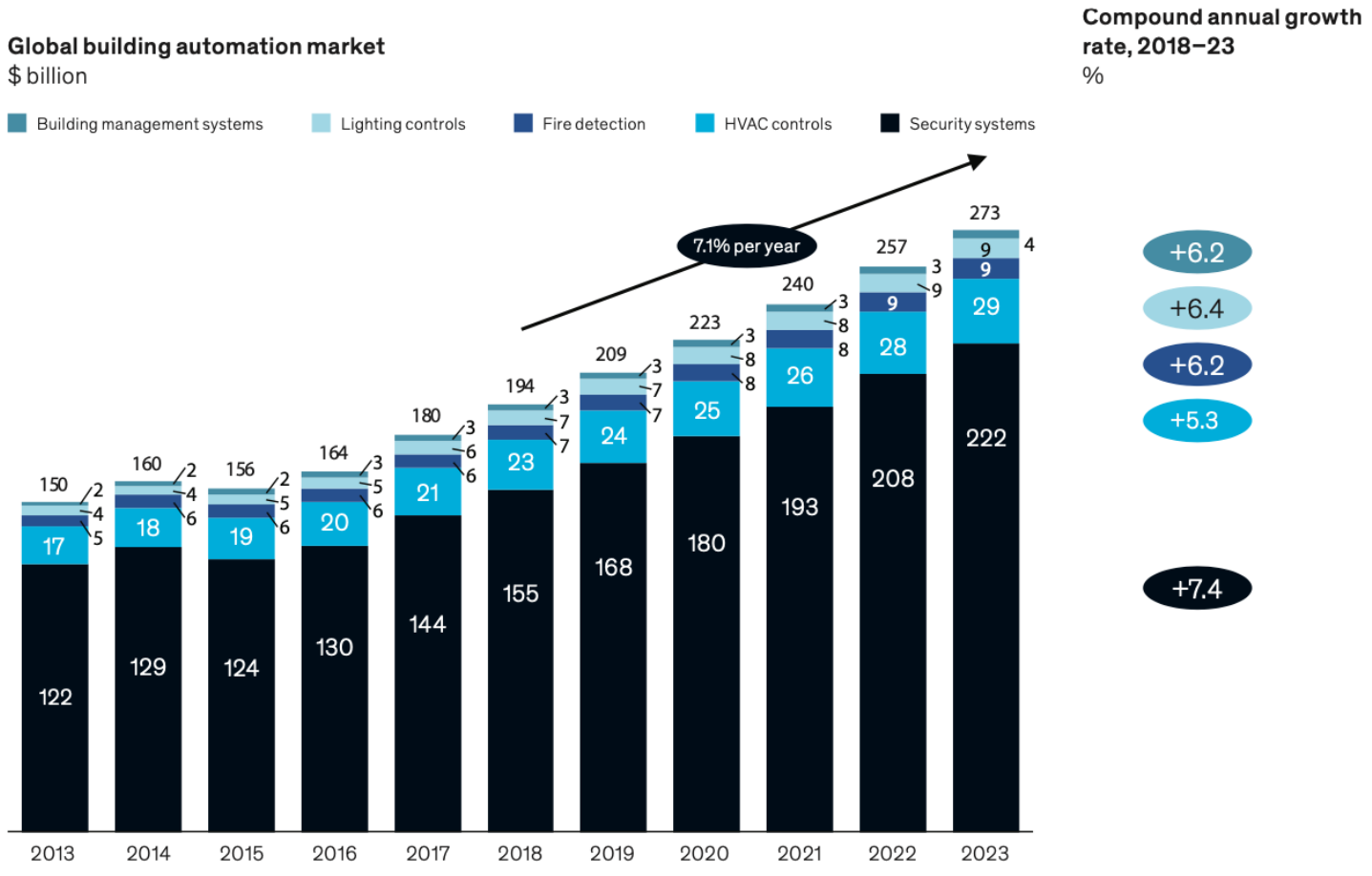
5. NPV of SGD8.5M with funding gap of SGD252K

Appendix

Total Addressable Market

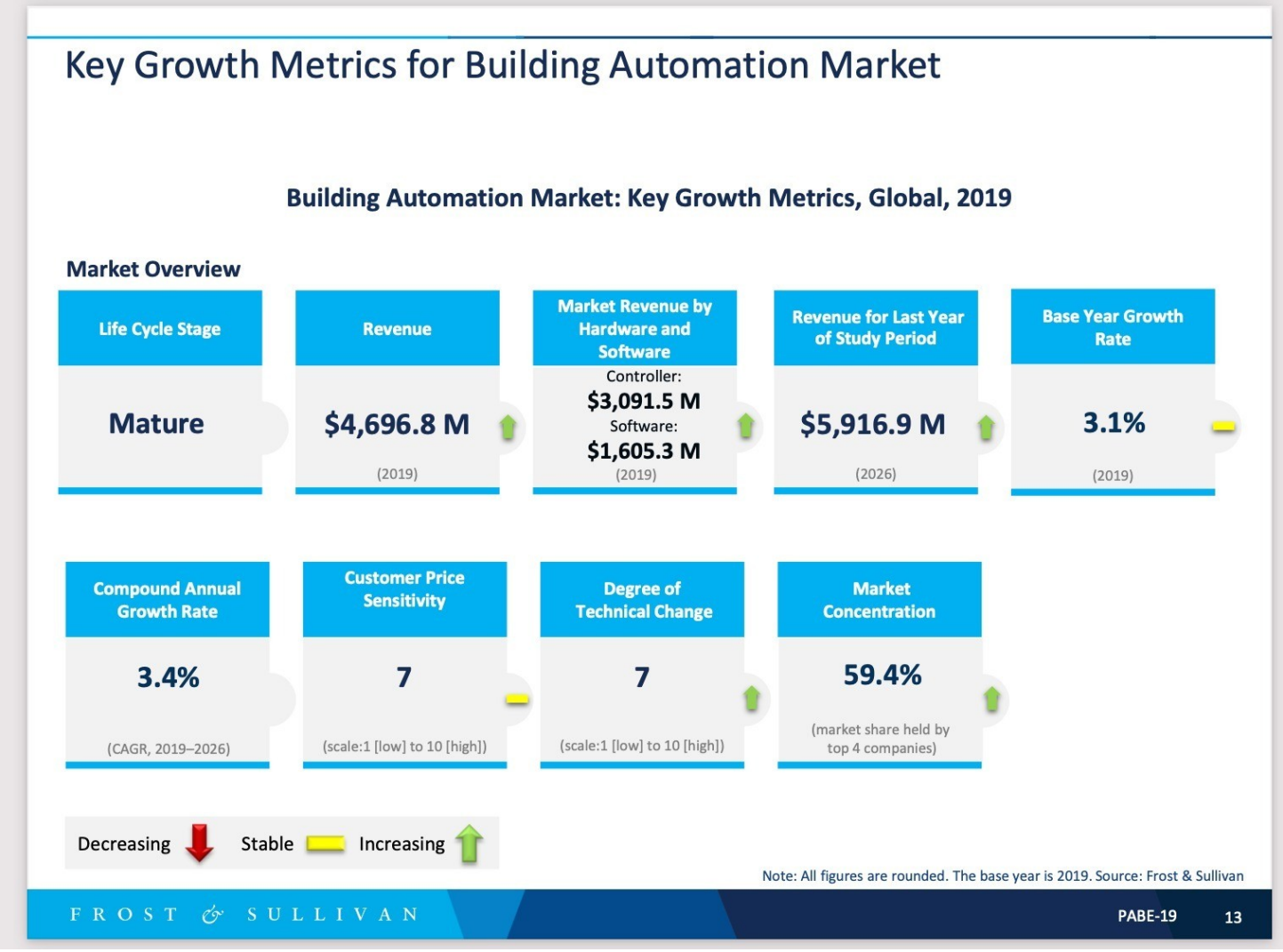
Exhibit 21

Building automation is projected to accelerate rapidly, particularly in security systems

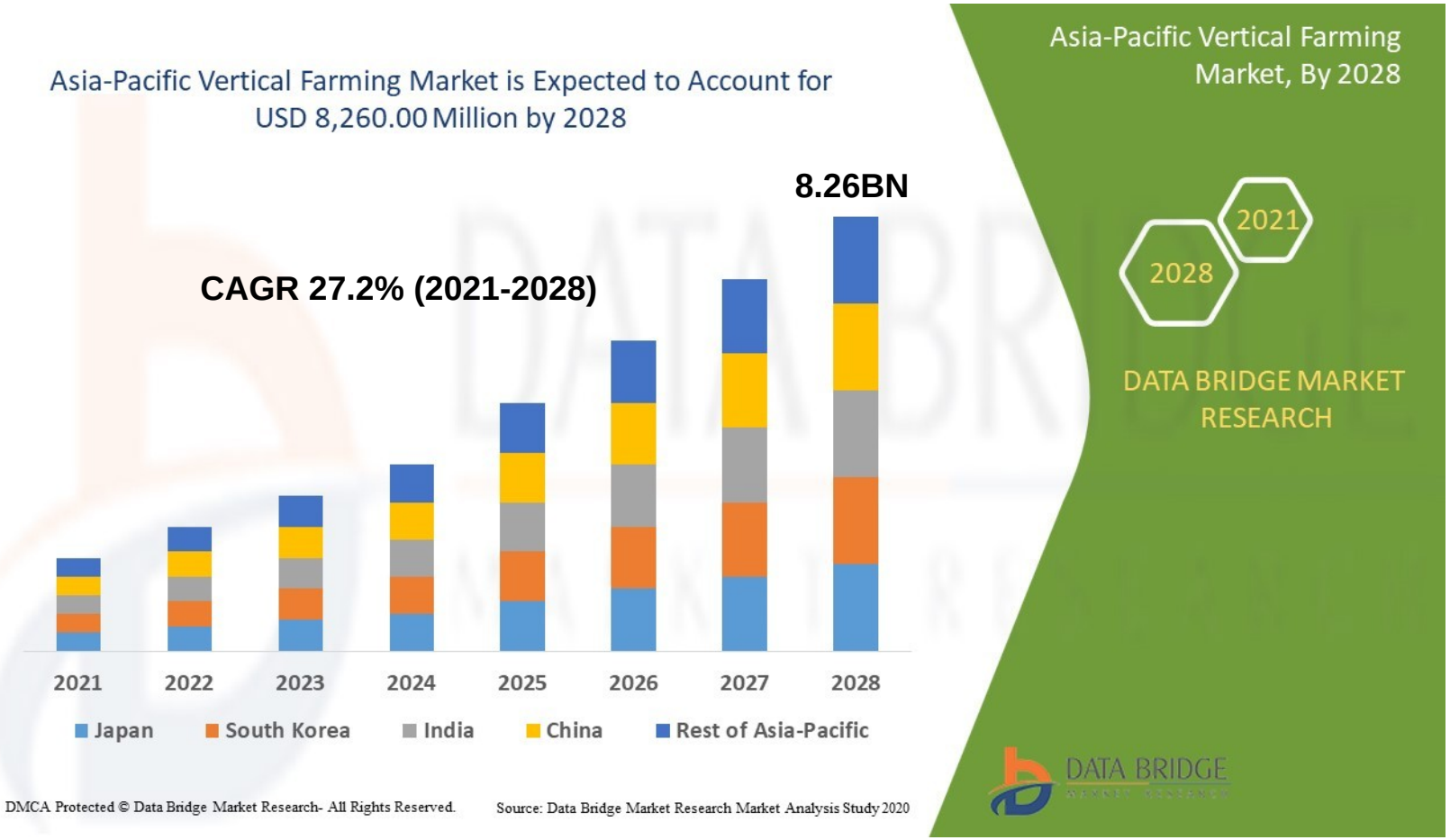


Source: Frost & Sullivan; IHS; Navigant

Serviceable Addressable Market



Serviceable Obtainable Market



DMCA Protected © Data Bridge Market Research- All Rights Reserved.

Source: Data Bridge Market Research Market Analysis Study 2020

Value-based Pricing

Source: Agfunder	Commercial Green House	Vertical Farming
Farm Size (SQFT)	280,000	67,000
Output (lbs)	2,000,000	2,000,000
Upfront Costs (USD/lbs)	0.61	1.3
Growing Costs (USD/lbs)		
Employee	1.1	1.57
Operating	0.42	0.42
Delivery Cost (USD/lbs)	0.2	0.2
Total Cost (USD/lbs)	2.33	3.07
Cost Differential (USD)		0.74
Cost Differential (%)		**24.10%**

1. We want to ensure price parity between operating a vertical farm and traditional greenhouse farms, in order to do that we need to bring price down by **24%**

	Min	Max
Cost of Employee (SGD)	$3,000	$4,000
Target Value Increase/pax	$723	$964
Monthly Value/sq ft	$1.45	$1.93

2. Tagging the **24%** increase in value to human capital productivity would mean monthly value derived/sqft to be around $1.45-$1.93

SinGrow		
Size (Sq Ft)	500	
Output (kg/harvest)	200	
No. of harvest/year	9	
Output (kg/year)	1800	
Weight of Punnet (kg)	0.2	
No. of Punnets/year	9000	
	Min	**Max**
Price/punnet	$9	$12
Annual Revenue	$81,000	$108,000
Monthly Revenue/Sqft	$13.50	$18

3. Monthly revenue/sqft from SinGrow's sales of strawberries is $13.50-$18/sqft

	Expected Value (%)	Min	Max
Energy Savings			
Decrease in utility costs	25%	$0.02	$0.02
Improved Crop Yield			
Increase in output	25%	$3.38	$5
Operational Efficiency			
Savings in man hours	25%	$0.39	$0.39
		$3.79	**$4.91**

4. Expected Value derived to be around $3.79-$4.91

Financial Model

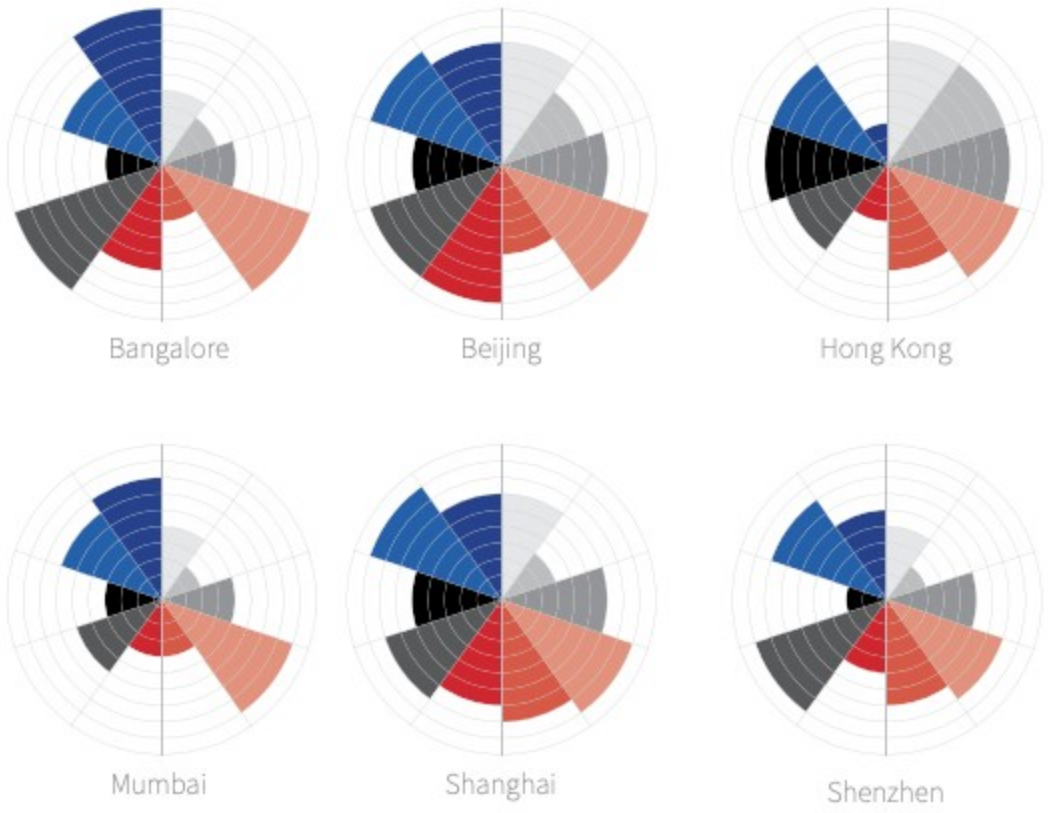
	Y1	Y2	Y3	Y4	Y5
Number of Customers managed	14	45	141	354	566
Area of Facility Size Managed (Sq Ft)	7,133	22,386	70,256	176,917	283,249
Ambient					
Revenue	$54,251	$488,319	$1,553,298	$3,961,237	$6,738,742
COGS	($14,597)	($107,885)	($262,720)	($448,786)	($561,349)
Gross Profit	$39,654	$380,435	$1,290,577	$3,512,451	$6,177,393
Profit Margin	73.1%	77.9%	83.1%	88.7%	91.7%
Operating Expense	($195,120)	($425,732)	($540,026)	($695,930)	($891,664)
P&L	($155,466)	($45,297)	$750,552	$2,816,521	$5,285,729
	($155,466)	($200,764)	$549,788	$3,366,309	$8,652,038
Discount Rate	8%				
Net Present Value	$8,483,135				

Use of GRIP Fund

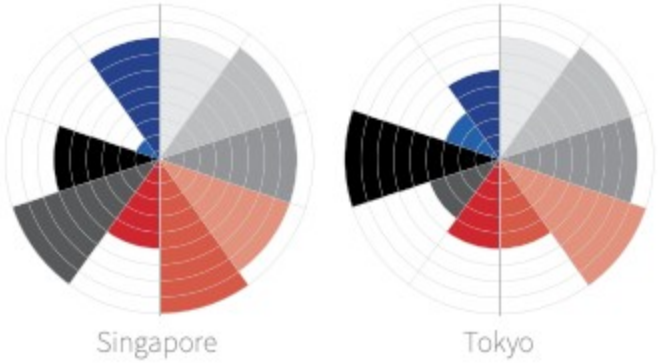
(S$)	Jul 2021	Aug 2021	Sep 2021	Oct 2021	Nov 2021	Dec 2021	Jan 2021	Feb 2021
Revenue								
Sensor Revenue	$8,571	$857	$943	$1,037	$1,141	$1,255	$1,380	$1,518
Integration Revenue	$0	$0	$0	$0	$0	$0	$0	$0
Software Revenue	$0	$0	$0	$0	$0	$0	$4,872	$5,359
	$0	$0	$0	$0	$0	$0	$0	$0
Data Insights Revenue	$0	$0	$0	$0	$0	$0	$0	$0
Total Revenue	**$8,571**	**$857**	**$943**	**$1,037**	**$1,141**	**$1,255**	**$6,252**	**$6,877**
Cost of Sensors	$8,057	$806	$886	$975	$1,072	$1,180	$1,159	$1,275
Cost of Integration	$0	$0	$0	$0	$0	$0	$0	$0
Cost of Maintenance	$0	$0	$0	$0	$0	$0	$0	$0
Total Cost of Goods Sold	**$8,057**	**$806**	**$886**	**$975**	**$1,072**	**$1,180**	**$1,159**	**$1,275**
Gross Profit	**$514**	**$51**	**$57**	**$62**	**$68**	**$75**	**$5,093**	**$5,602**
Operating Expense								
Rent	$0	$0	$0	$0	$0	$0	$0	$0
Tech Development	$4,500	$4,500	$4,500	$4,500	$4,500	$4,500	$6,000	$6,000
Salary	$6,000	$6,000	$6,000	$6,000	$6,000	$6,000	$12,000	$12,000
Others	$1,000	$1,000	$1,000	$1,000	$1,000	$1,000	$1,000	$1,000
Total Operating Expense	**($11,500)**	**($11,500)**	**($11,500)**	**($11,500)**	**($11,500)**	**($11,500)**	**($19,000)**	**($19,000)**
Profit & Loss	**($10,985.75)**	**($11,448.58)**	**($11,443.43)**	**($11,437.78)**	**($11,431.55)**	**($11,424.71)**	**($13,907.35)**	**($13,398.09)**
	($10,985.75)	($22,434.33)	($33,877.76)	($45,315.53)	($56,747.09)	($68,171.80)	($82,079.15)	($95,477.24)

Go-to Market Strategy

Future Asian agri-food tech investment and innovation hubs — relative strengths



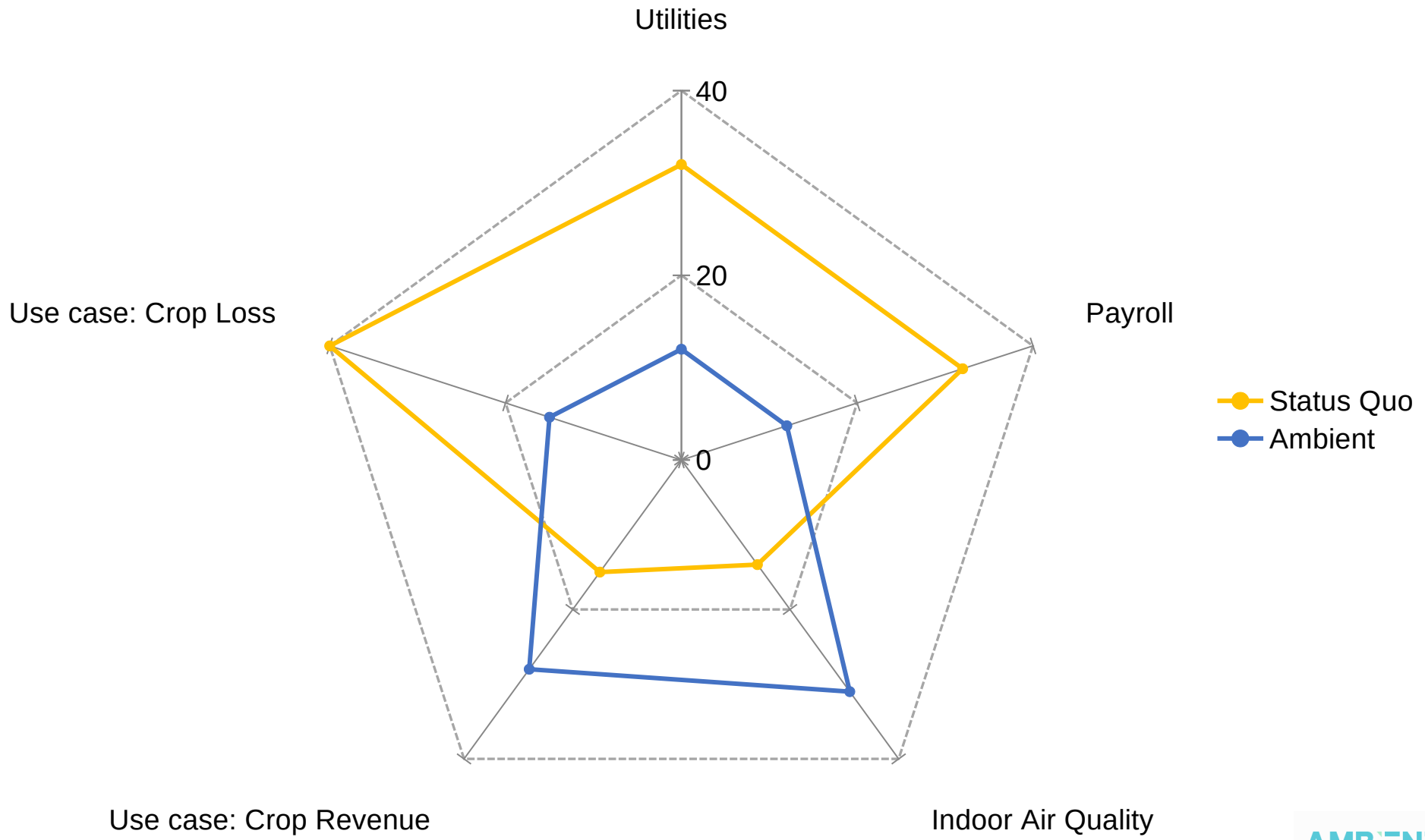
Number of Farms in Asia			
Country	Date	Type of Farms	Number
Singapore	2020	Leafy Vegetables	87
Hong Kong	2019	Organic Farms	322
Japan	2019	Indoor Farms	200
China	2020	PFAL	200



Sources: Capital IQ, Crunchbase, Economist Intelligence Unit, Eden Strategy Institute, INSEAD, Martin Prosperity Institute, Pitchbook, Quacquarelli Symonds, World Bank, PwC analysis

- Quality and engagement of academic institutions
- Talent competitiveness
- Quality of working and living environment
- Start-up ecosystem and entrepreneur culture
- Government initiatives supporting innovation and investment
- Existing presence of venture capital activities
- Technology availability
- Existing presence of large agri-food businesses
- Local market for agri-food technology
- Current levels of venture capital investment in agri-food tech businesses

Problem: Multi-objective Optimization

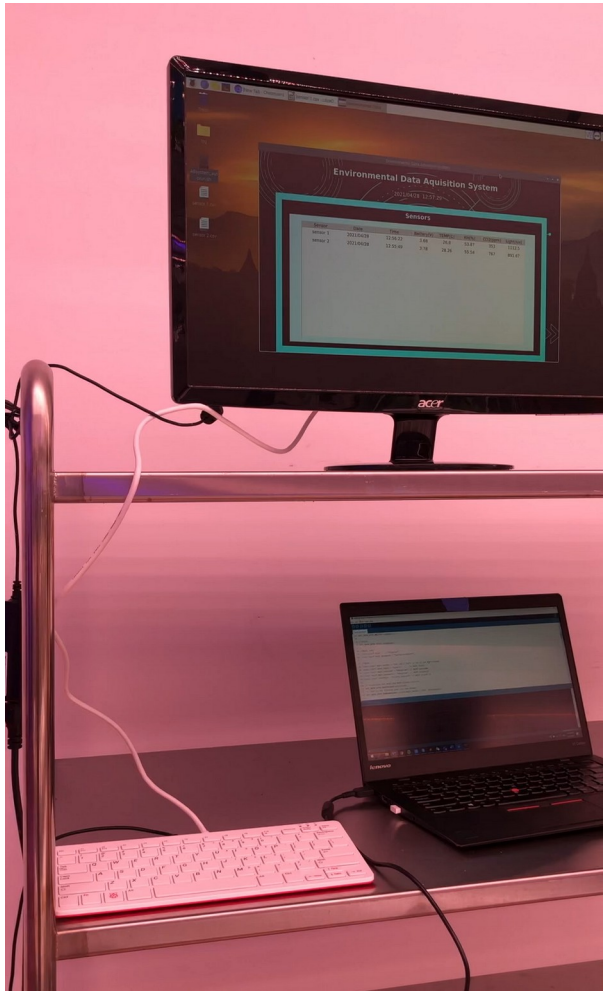


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Proof-of-Concept
MVP
Pilot
Outcomes

1. Collected climate data successfully
2. Identified gap between current and optimal
3. Applied s to determine optimal
4. Provided recommendations





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Solution Targets Quantified

	Parameter	Actual	Optimal	Sample Action
Current POC	Air Temperature	28 C / 20 C	25 C / 12 C	Decrease temperature setpoint Increase ventilation
	Relative Humidity	55% to 70%	60% to 75%	Install fogging machines
	CO2	300 to 700 ppm	>330 ppm 500 to 1200 ppm	Decrease/increase ventilation (depends on CO2 concentrations outside) Install CO2 regulator
	Spectrum Light	2300 lux	5000+ lux	Adjust non-farming lights Increase light intensity Increase natural light

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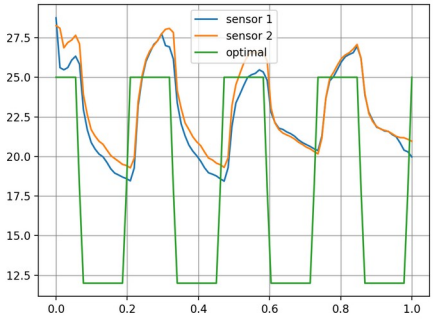
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Full-scale Plot	Soil Temperature	MVP 2.0		
	Soil Moisture			
	pH level			
	Electrical conductivity (EC_w)			
	Crop specific parameters			

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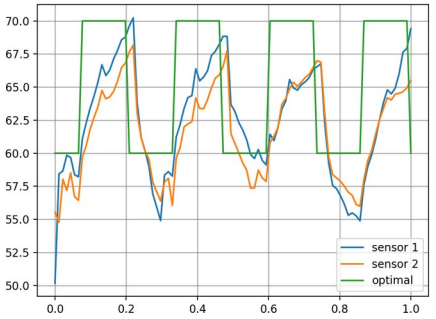
Correlation Analysis

Temperature



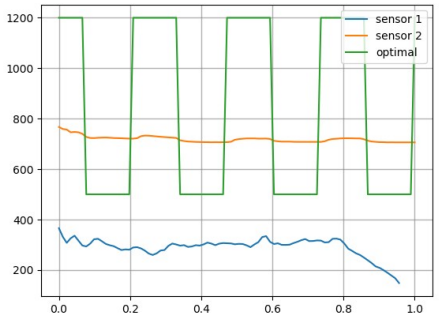
Leaf and petiole growth
Brighter leaf color
Greater sucrose
More biomass at the roots

Relative Humidity



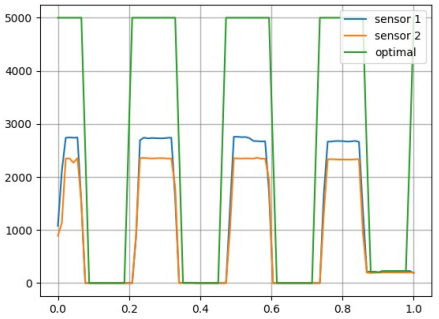
Avoid tip-burn and calyx burn
Avoid fungi
Avoid powdery mildew
Avoid leaf spot (bacterial blight)

CO2



Increased photosynthesis
Earlier flowering
Higher fruit yields
Improved stem strength and flower size

Light Intensity



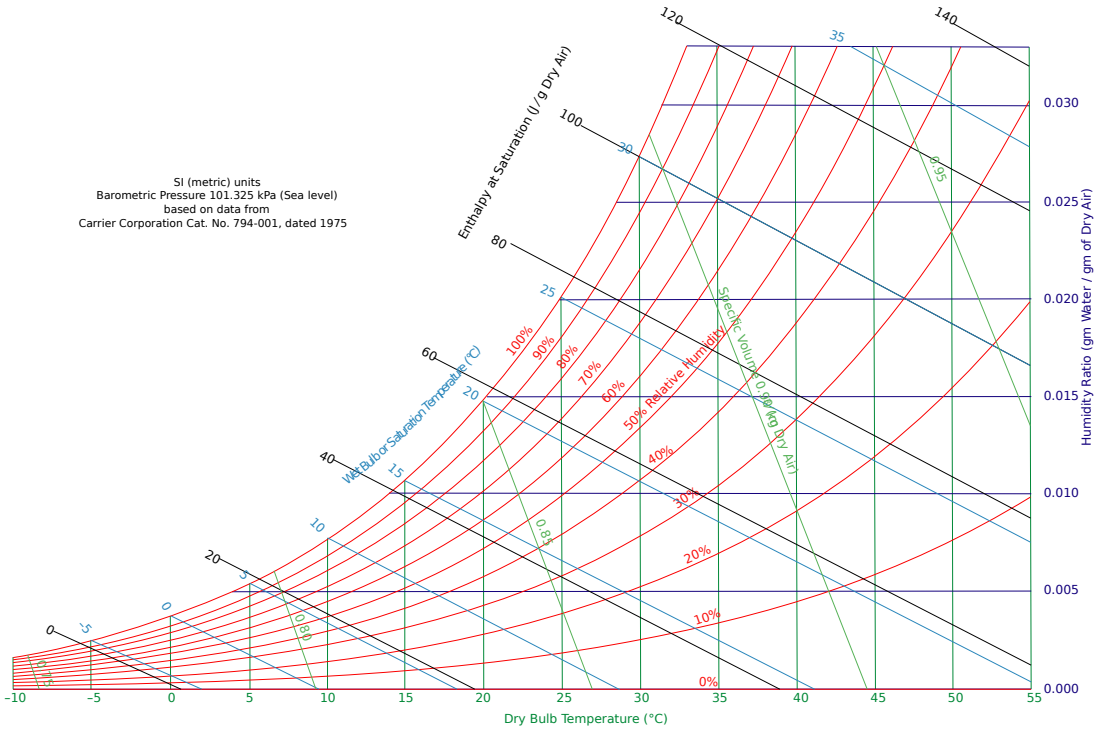
Increased photosynthesis

Temperatures after bloom affect plant growth and fruit quality of strawberry, , 2000
Crop Guide: Growing Strawberry, , 2004
Carbon Dioxide In Greenhouses, Ministry of Agriculture, Food and Rural Affairs, 2021
Carbon dioxide (CO2) applications in indoor growing,
Temperature And Co2 Concentration On Photosynthesis And Growth Of In Vitro, International Society for Horticultural Science, 1989

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#1: Interactions

- Psychometrics: temperature and relative humidity
- Light increases temperature and decreases humidity
- Decreasing temperature decreases CO2 concentrations in the air
- Increased ventilation introduces hot humid air from outside that needs to be cooled, purified, e hed, and re-humidified → very energy intensive → very expensive
- Safety: CO2 is needed by the plants, but human workers need oxygen to work indoors
- Photosynthesis process: lights and CO2 interact (both need to be high or both need to be low)
- Infiltration/exfiltration: porous building materials



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#2: Equipment

- Building equipment is often outdated
- AC controls are limited: only allows setting temperature and fan speed (not granular enough)
- AC controls cannot be managed remotely
- Need additional or custom equipment based on plant type, growing stage, and outdoor climate

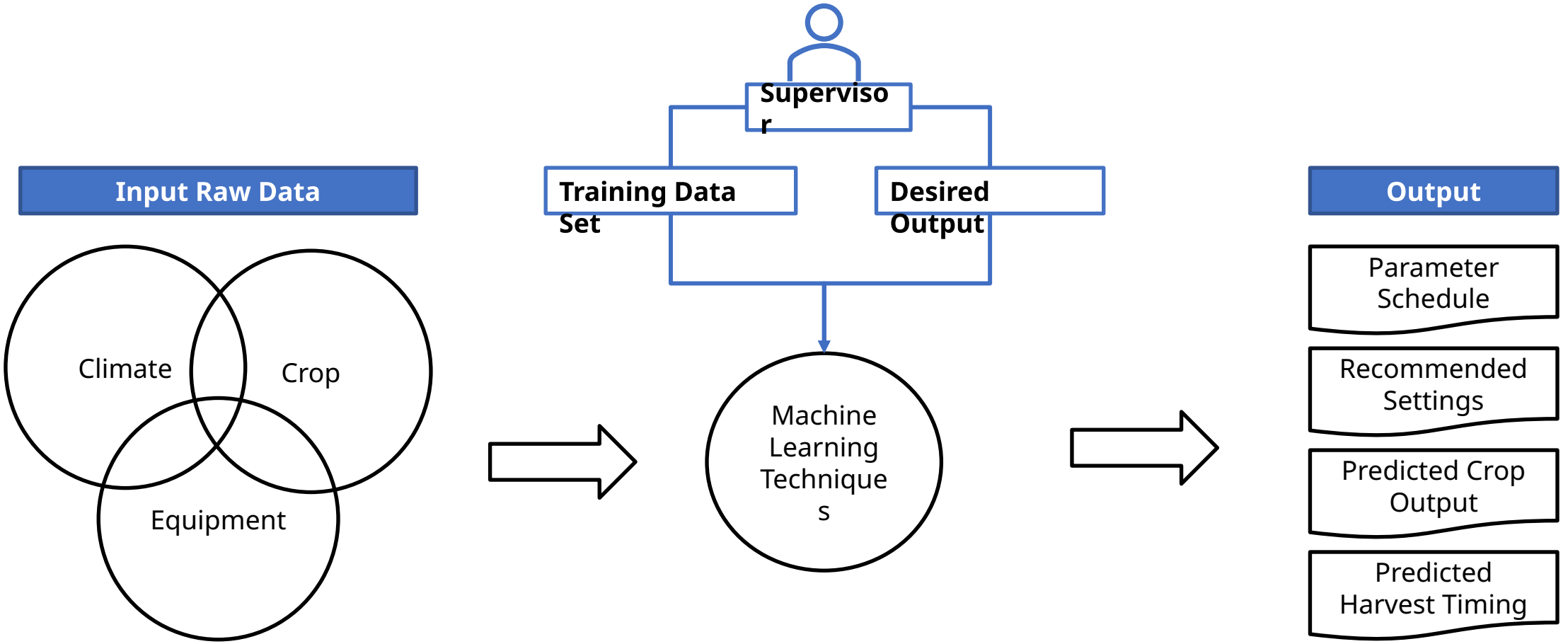






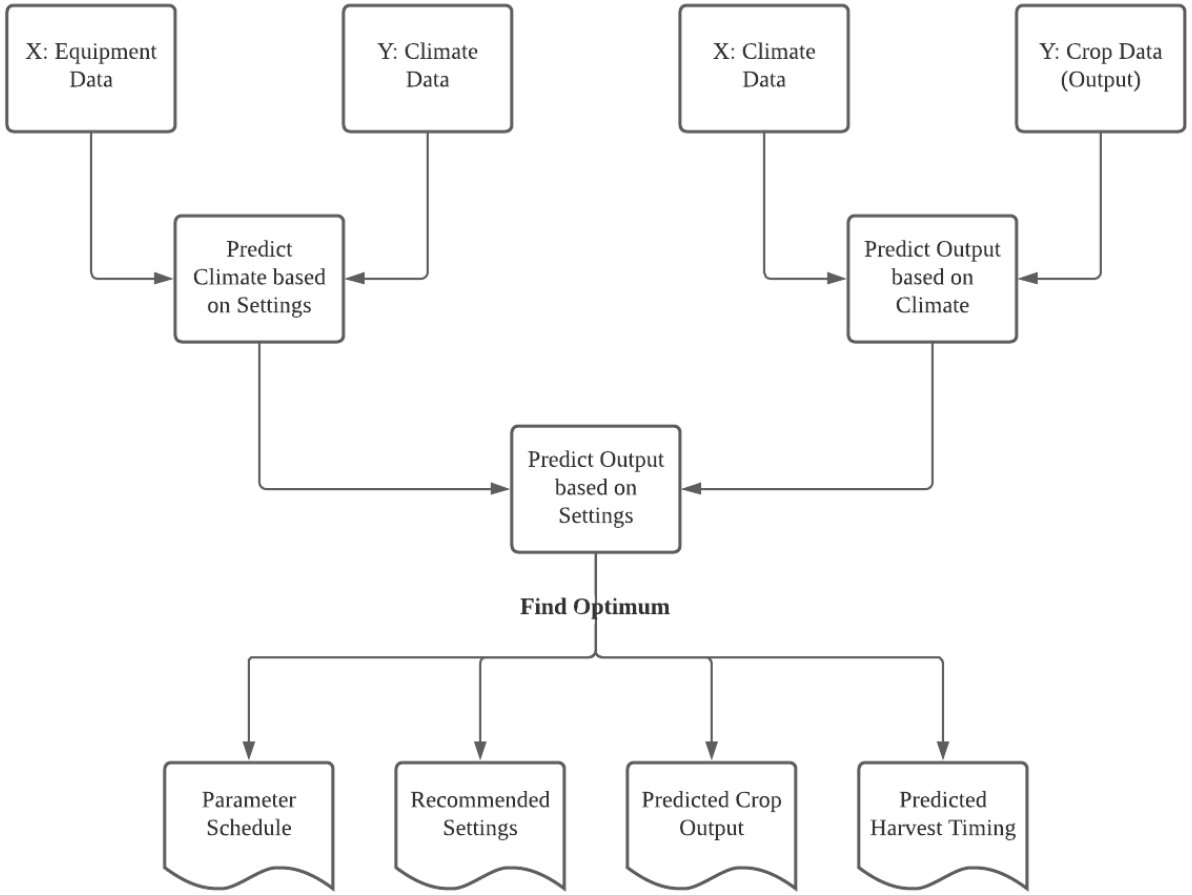
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Ambient's Machine Learning Optimization Model



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Supervised Learning: Three Stages



Notes:

- Non-climate parameters are already automated and controlled via dedicated systems (i.e., LED lighting, nutrition, irrigation, etc.)
- How Crop Data (output) is measured will vary based on farm type → it will impact data collected
- Product is agnostic to how crop output is defined or measured (assuming climate is an important input)
- More frequent data collection will increase the training of the feedback loop (e.g., providing crop data daily will allow quicker calibration than providing data at the end of each harvest)
- We can work with the farmer to provide the necessary data acquisition for crop quality (e.g., image recognition) and platform for training (e.g., optimization features)

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Patent Classification

A	HUMAN NECESSITIES
B	PERFORMING OPERATIONS; TRANSPORTING
C	CHEMISTRY; METALLURGY
D	TEXTILES; PAPER
E	FIXED CONSTRUCTIONS
F	MECHANICAL ENGINEERING; LIGHTING; HEATING; WEAPONS; BLASTING
G	**PHYSICS** **G06:** COMPUTING; CALCULATING OR COUNTING **G06N:** COMPUTER SYSTEMS BASED ON SPECIFIC COMPUTATIONAL MODELS **G06N 3/00**: Computer systems based on biological models **G06N 5/00**: Computer systems using knowledge-based models **G06N 7/00**: Computer systems based on specific mathematical models **G06N 20/00**: Machine learning
H	ELECTRICITY

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Productivity & Operational Efficiency

Ambient's optimization models show you how to create an optimal environment with:

- No need to manage sensors or raw data
- No need to manually adjust ACMV equipment early in the morning and late in the afternoon to accommodate plant growing schedule
- No need to manually adjust built-in (non-farming) lights
- No need for analysis of complex interactions between parameters
- No need to wait until seeing harvest results to fine-tune the parameters
- No need to deal with complex thermal inertia calculations
- No need for complex set-ups to perform farming experiments
- No need to spend time evaluating which mechanical equipment to buy (we recommend it)
- No need to worry about finding and selecting inter-operable systems (we shortlist them for you)

How much time can Ambient save you?

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Recommendations

Insight	Recommendation	Cost
No historical data	Measure and securely archive critical operational data for crop repeatability and insurance	Ambient Platform Tier 1
Variability between days	Monitor for longer period of time (entire harvest cycle)	Ambient Platform Tier 1
No utilities data	Monitor energy consumption	$5k-$10k hardware modification Ambient Platform Tier 1 & 2
High temperature	Decrease daytime and night-time temperature	$5k - $10k portable AC
Low CO2	CO2 regulator (increase CO2)	$1k - $2k CO2 regulator $500 monthly CO2 supply
Moisture timing	Introduce mist earlier in the day Reduce mist later in the day	Ambient Platform Tier 1
Plants have various optimal parameter schedules	Determine farm portfolio optimal (not just for one plant but entire farm)	Ambient Platform Tier 1 & 3

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